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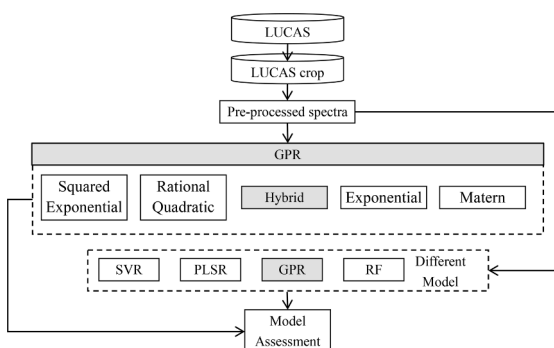
Estimation of soil organic carbon in LUCAS soil database using Vis-NIR spectroscopy based on hybrid kernel Gaussian process regression

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HIGHLIGHTS

- Estimating SOC using Vis-NIR absorption spectroscopy in large soil databases.
- Comparing the estimation performance of PLSR, SVR, RF, and GPR.
- Proposing a GPR model based on a hybrid kernel function.

GRAPHICAL ABSTRACT



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ABSTRACT

Soil Organic Carbon (SOC) is crucial for determining soil fertility and environmental quality. The problem with traditional SOC chemical analysis methods is that they are time-consuming and resource-intensive. In recent years, visible-near infrared (Vis-NIR) spectroscopy has been employed as an alternative method for SOC determination. However, when applied on a larger scale, the prediction accuracy of soil properties decreases due to the heterogeneity of samples. Therefore, this study compared and analyzed the performance of partial least squares regression (PLSR), support vector regression (SVR), random forest (RF), and gaussian process regression (GPR) in predicting SOC. On this basis, a GPR model based on a hybrid kernel function (HKF-GPR) was proposed for SOC prediction. This hybrid kernel function was designed according to the properties of single kernel functions and the characteristics of soil spectral data. Results indicate that in large soil spectral databases, the GPR model outperforms other models in estimating SOC. The HKF-GPR model achieved the best SOC estimation accuracy, with an R^2 of 0.7671, RMSE of 5.2934 g/kg, RPD of 2.0721, and RPIQ of 2.5789. Compared to other regression models, the HKF-GPR model proposed in this paper offers broader applicability and superior performance, enabling SOC estimation in large soil spectral libraries.

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1. Introduction

Soil degradation is a critical issue globally, with significant implications for climate change and food security [14]. Particularly, soil organic carbon (SOC) is an essential soil attribute, playing a pivotal role in promoting the formation of soil aggregates, influencing the quality of arable land, and facilitating the global carbon cycle [19]. Accurate assessment of SOC is therefore vital for monitoring global climate change, constructing carbon cycle models, and developing sustainable agriculture.

The traditional methods for determining SOC mainly include potassium dichromate oxidation with external heating, potassium dichromate oxidation with spectrophotometry, total organic carbon analyzer method, and elemental analyzer method. Dematte et al. (2001) measured the organic matter content of 39 soil samples using the colorimetric method. Nanni and Demattè [17] measured the organic matter content of 368 soil samples using the Walkley-Black method. However, considering the high labor intensity and cost associated with traditional soil analysis techniques, identifying convenient and economically feasible analysis methods becomes crucial [27]. Consequently, the introduction of visible-near infrared (Vis-NIR) spectroscopy technologies represents a significant advancement, providing a viable alternative for the accurate quantification of soil properties [9]. Soil spectroscopy analysis rests on the foundational premise that there exists a linear correlation between the concentrations of specific soil attributes and the amalgamation of characteristic spectral absorptions Bellon-Maurel and McBratney [3]. This absorption is primarily induced by the harmonics and combination frequencies resulting from the fundamental vibrational modes of molecular functional groups, notably the hydroxyl (OH) group. The specific wavelengths associated with the harmonics and combination bands of each functional group facilitate the accurate discrimination of various substances [6].

The spectrum of soil reflectance features overlapping broad and subdued absorption characteristics [22]. To decipher quantitative data from the minimal quantities of soil components, a variety of mathematical modeling techniques are employed (for comparison, see [23]). Among these, partial least squares regression (PLSR) stands out as a frequently utilized method [21]. PLSR is particularly effective in handling a substantial number of interrelated predictor variables, a common scenario in diffuse reflectance spectroscopy [22]. Moura-Bueno et al. [16] compared the predictive performance of six preprocessing techniques and four calibration models on the organic carbon content of 841 subtropical soil samples. The best preprocessing technique was the first derivative of SG smoothing, and the best calibration model was PLSR.

In 2015, the Land Use/Cover Area frame Survey (LUCAS) was conducted across the 28 member states of the European Union with the aim of establishing a coherent pan-European database of physical and chemical topsoil properties. In countries sampled in 2009 and 2012, 90 % of sampling points were retained, while the remaining 10 % were replaced with new sampling points Jones et al. [10]. After excluding unidentifiable samples, the LUCAS 2015 soil dataset comprises information on 21,859 soil and agricultural environmental data points. Several physical and chemical parameters of the soil samples were analyzed, and the soil samples were scanned with a visible-near infrared spectrometer in separate laboratories. A notable advantage of the LUCAS database is that all physical, chemical, and spectral measurements were conducted using uniform standards and protocols [24].

Nevertheless, at broader scales (from regional to global) the predictive accuracy of regression models diminishes. This decline is primarily attributed to the escalating spatial heterogeneity of soil properties, presenting a significant challenge to the generalization capabilities of these models [1]. The prevalence of nonlinear relationships within large databases significantly diminishes the efficacy of PLSR models. The random forest (RF) algorithm, an ensemble approach that compiles predictions from multiple decision trees to produce a final

forecast, demonstrates enhanced capability in navigating nonlinear interactions among variables and mitigating overfitting. Accordingly, it has been extensively utilized in current soil hyperspectral research [8]. Meng et al. [15] employed a stratification strategy and the RF algorithm to enhance the predictive accuracy of SOC content, achieving a reduction in RMSE of 4.88 g kg^{-1} and an increase in R^2 by 0.32. Nonetheless, the RF algorithm exhibits a 'step-like' behavior in predicting continuous variables, which may reduce its efficacy in analyzing hyperspectral data. Furthermore, its method of adapting to the particular distribution features of data—through modifications in the depth and quantity of decision trees—does not afford the requisite flexibility for managing extensive and intricate soil databases [29].

In response to the deficiencies observed in the RF algorithm, this study adopts Gaussian process regression (GPR) with kernel functions to formulate a soil hyperspectral estimation model. The implementation of kernel functions within GPR effectively resolves computational challenges in high-dimensional feature spaces, showcasing notable efficacy in the system identification of dynamic ship models. Xue et al. [31] developed an input noise Gaussian Process model, wherein input uncertainty is systematically apportioned to the model's output via Taylor approximation. This approach significantly improves the precision and reliability of the ship system identification framework. Liu et al. [12], Liu et al. [13] developed an innovative system identification technique for forecasting ship motions, employing GPR integrated with Wavelet Threshold (WT) denoising to specifically address noise reduction. GPR, as a kernel-based method, naturally accommodates the smoothness and continuity of data, which is particularly beneficial in the analysis of spectral data. However, there have been few reports on the use of kernel-based GPR for estimating SOC. Moreover, beyond the commonly utilized Squared Exponential (SE) kernel function, other kernel functions exist, such as the Matern kernel, Exponential kernel, and Rational Quadratic (RQ) kernel. Nonetheless, these singular kernel functions may not fully capture the spectral heterogeneity present within large soil databases.

In response to the shortcomings of existing soil hyperspectral models, our research focuses on the LUCAS 2015 soil database. We compared and evaluated the model performance of PLSR, support vector regression (SVR), RF, and GPR in estimating SOC content using Vis-NIR spectra, thereby confirming the effectiveness of GPR. Furthermore, we explored the impact of different kernel functions on GPR and introduced a model based on a hybrid kernel function (HKF-GPR). This model combines high smoothness with improved adaptability to rough processes, thereby enhancing the precision of SOC estimation.

2. Materials and methods

2.1. LUCAS soil database

The LUCAS topsoil database represents the most extensive and consistent soil database across the European continent to date, providing support for policy specification Jones et al. [10]. This database encompasses 12 different soil properties, including SOC and spectral measurements within the Vis-NIR to Short Wave Infrared (VNIR-SWIR) range [23]. Total carbon is measured by dry combustion at 900°C using the VarioMax CN analyzer (Elementar Analysensysteme GmbH, Germany). The SOC content is then derived by subtracting the carbonate content (measured according to ISO 10693:1995) from the total carbon. Soil Vis-NIR absorbance is measured in the 400–2500 nm wavelength range using the FOSS XDS Rapid Content Analyzer (FOSS NIRSystems, Denmark), with a spectral resolution of 2 nm and a spectral data interval of 0.5 nm, yielding a total of 4200 wavelengths.

In alignment with the land use and cover classifications of LUCAS 2015, we have delimited our study to the agricultural sectors within the database, comprising 12,817 soil samples. Our focus on agricultural areas is due to their periodic lack of vegetation post-harvest, a feature that makes them particularly suitable for soil property estimation through remote sensing from aerospace platforms.

2.2. Preprocessing

To eliminate redundant information among spectral data, the spectral data intervals were resampled to 2 nm, reducing the number of spectral bands to 1050. Numerous studies have demonstrated that first derivative transformation can decrease or eliminate noise, enhance the correlation between spectra and SOC, and improve model prediction accuracy [26,1]. We also applied the first derivative transformation as a spectral preprocessing method. Stevens et al. [23] identified instrument artifacts within the 400–500 nm spectral range. Hence, spectral bands between 400–498 nm, encompassing 50 spectral bands, were excluded in this study, leaving 999 spectral bands after preprocessing.

To enhance the predictive performance of our model, we applied a logarithmic transformation to the SOC content during model calibration [28]. For performance evaluation, the estimated SOC values were back-transformed.

In this research, the Kennard-Stone technique [11] divided the LUCAS 2015 agricultural subset into a training set (70 %) and a test set (30 %). The test set was exclusively used for evaluating model performance, while the training set was employed for calibrating regression models and optimizing hyperparameters.

2.3. GPR models with a hybrid kernel function

2.3.1. GPR

GPR constitutes a non-parametric modeling framework that leverages Gaussian process priors for the regression analysis of datasets. This methodology inherently encompasses both regression residuals and the Gaussian process prior as its foundational elements [7]. Central to the GPR paradigm is the presupposition that the empirical data adhere to a Gaussian distribution. Employing Bayesian inference, GPR facilitates the prediction of unknown data points through an analytical exploration of the probabilistic distribution characteristics Williams and Rasmussen [30].

For any given training sample dataset $D:(x,y)$, let $f(x_i) = y_i$, and assume that both follow a multivariate joint Gaussian distribution, then:

$$\begin{cases} f(X) = [f(x_1), f(x_2), \dots, f(x_n)] - GP\left[\mu(x), \sum(x)\right] \\ Y = f(X) + \varepsilon - N\left[\mu(x), \sum(x) + \sigma^2 I\right] \end{cases} \quad (1)$$

$$\mu(x) = [\mu(x_1), \mu(x_2), \dots, \mu(x_n)] \quad (2)$$

$$\begin{bmatrix} K(x_1, x_1) & K(x_1, x_2) & \dots & K(x_1, x_n) \\ K(x_2, x_1) & K(x_2, x_2) & \dots & K(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ K(x_n, x_1) & K(x_n, x_2) & \dots & K(x_n, x_n) \end{bmatrix} \quad (3)$$

in the equation, μ represents the vector composed of the means of $f(x_1), f(x_2), \dots, f(x_n)$; $\sum(x)$ denotes its covariance matrix; K denotes the kernel function. In this paper, the input variable n is set to the number of spectral bands, where $x_1, x_2, \dots, x_n$ are the absorbance values for each band; Y represents the SOC content.

Define the set of x_i to be predicted as $X^* = (x_1^*, x_2^*, \dots, x_n^*)$. The corresponding predicted values are:

$$y^* = f(x^*) + \varepsilon \quad (4)$$

GPR begins by calculating the joint probability distribution $f \sim N(\mu, K)$ among the samples in the data set. Then, the posterior probability distribution of f^* is calculated based on the prior probability distributions $f^* \sim N(\mu^*, K^*)$ and $f \sim N(\mu, K)$ of f^* . Jointly with the initial data, the following equation is derived:

$$\begin{bmatrix} Y \\ f(x^*) \end{bmatrix} = N\left(\begin{bmatrix} \mu(x) \\ \mu(x^*) \end{bmatrix}, \begin{bmatrix} K + \sigma^2 I & K(x, x^*) \\ K(x^*, x) & K(x^*, x^*) \end{bmatrix}\right) \quad (5)$$

where: $K(x, x^*) = [K(x_1, x_1^*), K(x_2, x_1^*), \dots, K(x_n, x_1^*)]$. Calculating the probability distribution of Eq. (5) yields Eq. (6):

$$\begin{cases} P(f(x^*)|Y, X, x^*) = N(\mu^*, \sigma_x^2) \\ \mu^* = \mu(x^*) + k(x^*, x)(K + \sigma^2 I)^{-1}[Y - \mu(x)] \\ \sigma_x^2 = k(x^*, x) - k(x^*, x)(K + \sigma^2 I)^{-1}k(x, x^*) \end{cases} \quad (6)$$

Based on the principles of GPR outlined above, a model for estimating SOC content is established.

2.3.2. Design of the hybrid kernel function

The essence of GPR lies in its ability to flexibly adapt to varying levels of data complexity by adjusting the kernel function and its parameters, thus enabling effective modeling of both complex and simple data patterns. Consequently, the selection of an appropriate kernel function is paramount in GPR. The Squared Exponential (SE) kernel function is the most widely applied, along with the Rational Quadratic (RQ) kernel function, Exponential kernel function, and Matern kernel function. The expressions for these functions are as follows:

$$K_{SE}(x, x^*|\alpha, \beta) = \alpha^2 \exp\left(-\frac{r^2}{2\beta^2}\right) \quad (7)$$

$$K_{RQ}(x, x^*|\alpha, \beta, \gamma) = \alpha^2 \left(1 + \frac{r^2}{2\gamma\beta^2}\right)^{-\gamma} \quad (8)$$

$$K_E(x, x^*|\alpha, \beta) = \alpha^2 \exp\left(-\frac{r}{\beta}\right) \quad (9)$$

$$K_{Matern}(x, x^*|\alpha, \beta) = \alpha^2 \left(1 + \frac{\sqrt{3}r}{\beta}\right) \exp\left(-\frac{\sqrt{3}r}{\beta}\right) \quad (10)$$

where, r denotes the Euclidean distance between x and x^* , i.e., $r = \sqrt{(x - x^*)^T(x - x^*)}$; α denotes the standard deviation of the signal; β denotes the feature length scale; and γ denotes the positive scale mixing parameter.

The SE kernel function is infinitely differentiable, meaning it can produce very smooth functions. However, its sensitivity to points far from the training data may lead to overfitting. The RQ kernel function can be seen as a generalization of the SE kernel, introducing an additional parameter to control variations across different scales, offering greater flexibility but making the model more challenging to train and interpret. The Exponential kernel function is not infinitely differentiable, making it suitable for certain types of non-smooth processes, but its capability to capture complex relationships in data may be limited. The smoothness of the Matern kernel function lies between that of the SE and Exponential kernels, and it is also not infinitely differentiable.

To summarize, the combination of the SE and Matern kernels enhances the GPR model with both high smoothness and improved adaptability to rough processes. This flexibility is particularly beneficial for analyzing soil hyperspectral data characterized by both long-range smoothness and short-range fluctuations. The formula for the hybrid kernel function is:

$$K_{Hybrid}(x, x^*|\alpha, \beta) = a \bullet \alpha^2 \exp\left(-\frac{r^2}{2\beta^2}\right) + b \bullet \left(\alpha^2 \left(1 + \frac{\sqrt{3}r}{\beta}\right) \exp\left(-\frac{\sqrt{3}r}{\beta}\right)\right) \quad (11)$$

where, r denotes the Euclidean distance between x and x^* , i.e., $r = \sqrt{(x - x^*)^T(x - x^*)}$; α denotes the standard deviation of the signal; β denotes the feature length scale; a and b are weights, which, based on the characteristics of the soil spectral data, we set a to 5 and b to 1.

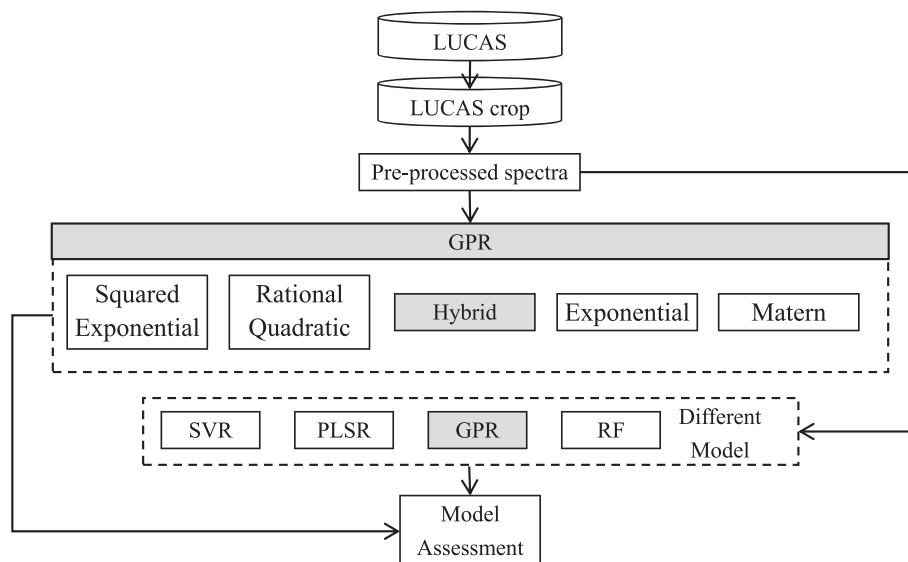


Fig. 1. Overall workflow diagram.

2.4. Other regression model

We compared and analyzed the model performance of SVR, PLSR, RF, and GPR to validate the effectiveness of the GPR model in estimating SOC.

PLSR integrates the advantages of Principal Component Analysis (PCA), Canonical Correlation Analysis, and Least Squares Regression, capable of performing regression when the number of samples is significantly less than the number of variables [21]. This method initially projects high-dimensional spectral information into a lower-dimensional space, aiming to maximize the covariance between spectral and soil component information. Subsequently, it establishes the relationship between spectral data and soil components in this reduced space. Unlike Principal Component Regression (PCR), PLSR extracts principal components from both the predictor matrix X and the response matrix Y , whereas PCA only extracts the principal components of the predictor matrix X . Therefore, the principal components in PLSR are often referred to as “Latent Variables (LVs)”.

SVR is a machine learning methodology grounded in statistical learning, representing a supervised learning approach aimed at minimizing empirical error and maximizing geometric margin. Breton and Lloyd [5]. Utilizing the principle of minimal structural risk and VC dimension theory, SVR is capable of identifying statistical laws with minimal structural risk in scenarios with limited sample sizes, thereby enhancing the model’s learning and predictive capabilities. Vapnik [25]. The core of the SVR model involves using kernel functions to map the target space into a high-dimensional feature space, rendering the samples linearly separable within this space. The model is optimized by maximizing the margin width and minimizing the total loss, thereby refining the prediction accuracy. Vapnik [25].

RF is a machine learning algorithm based on the Bagging ensemble technique, introduced by Breiman in 2001, and is comprised of multiple decision trees [4]. The fundamental principle of RF is to utilize the

Bootstrap sampling method to draw multiple samples from the input data with replacement. These sampled subsets are used to construct decision trees, with each leaf node randomly selecting features for training. Subsequently, the best split point within this feature set is identified to partition into left and right subtrees. Feature importance is estimated using out-of-bag data, facilitating variable selection and dimensionality reduction. Each decision tree yields a prediction outcome, and the final prediction is derived by averaging the outcomes across all decision trees.

2.5. Model assessment

The overall workflow of this study is illustrated in Fig. 1. Initially, the soil spectral data from the LUCAS 2015 agricultural subset are pre-processed. Subsequently, the performance of GPR using four different kernel functions for SOC estimation is compared. Based on this, a hybrid kernel function is proposed. Finally, the performance of different regression models in estimating SOC is compared.

To evaluate the predictive efficacy of various regression models and kernel functions, we employed a set of performance indicators, comparing laboratory-measured values against predictions derived from an independent test set. These indicators encompass the coefficient of determination (R^2), root mean square error (RMSE), the ratio of performance to deviation (RPD), and the ratio of performance to the interquartile range (RPIQ). Notably, RPD and RPIQ, recognized as relative analysis errors, facilitate the comparison of model performance across distinct studies. The computational simulations were conducted using MATLAB R2022a on an Intel® Core™ i5-8265U CPU @ 1.6 GHz with 16 GB RAM, under a Windows 10 64-bit operating system. All the code, training set, test set, and validation set in this paper were publicly available on Github at <https://github.com/Lbyyyyyyy/Hybrid-kernel-Gaussian-process-regression-Matlab>.

Table 1

Descriptive statistics of the agricultural subset SOC in the LUCAS database.

Dataset	SOC (g/kg)								
	N	Min	Max	Q25	Median	Q75	Mean	Std	Skew
Total	12817	0.1	534.8	11.20	16.80	26.72	25.12	34.97	7.40
Train	8971	0.1	534.8	11.40	17.60	29.27	27.96	40.74	2.43
Test	3846	0.1	147.7	11.00	15.25	22.10	18.48	11.94	2.73

Q25: the first quartile split of the lowest 25%; Q75: the third quartile split of the highest 75%.

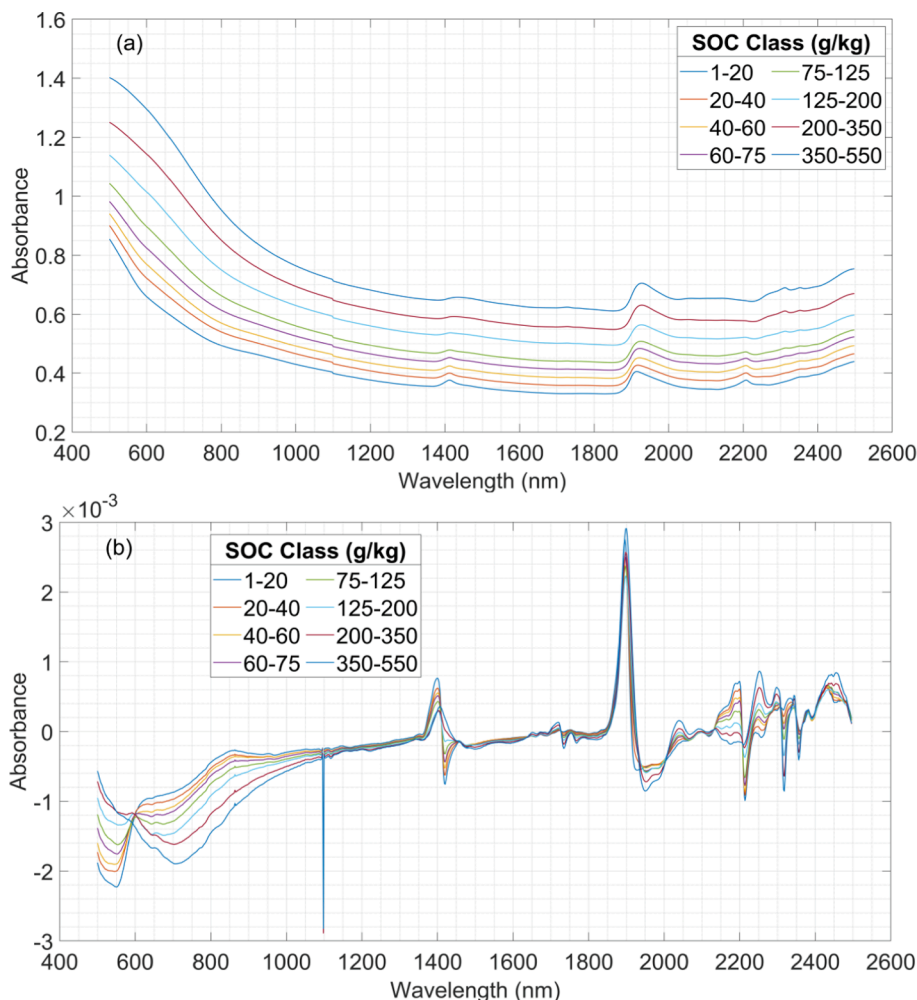


Fig. 2. Absorption spectra of different levels of SOC content in the LUCAS agricultural subset: (a) original spectra and (b) first derivative spectra.

3. Results and discussion

3.1. Spectral characteristics and SOC distribution

The agricultural subset from LUCAS indicates that the range of SOC content spans from 0.1 to 534.8 g/kg, with an average of 25.1 g/kg. The range of clay content is between 2 % and 62 %, averaging at 21 %. The CaCO_3 content varies from 0 to 976 g/kg, with an average value of 68.2 g/kg. Table 1 provides the descriptive statistics for SOC, revealing that the data for all datasets exhibit a skewed distribution. The skewness of the SOC values is similar for both the train and test datasets. However, the range of SOC content is wider in the train set, leading to a significantly higher standard deviation compared to the test set, indicating a broader dispersion of SOC values within the train dataset.

Spectral absorbance varies significantly with SOC content and mineral composition, as illustrated in Fig. 2. As depicted in Fig. 2a, spectral absorbance increases with the SOC content due to the darker color of soils rich in organic carbon [22]. The difference in absorbance becomes more pronounced with higher SOC contents in the Vis region compared to the NIR region [18]. Stenberg et al. [22] also observed a decrease in absorption rates of organic soils in the NIR region. The first derivative of the absorption spectrum (Fig. 2b) reveals a strong correlation between SOC content and absorption in the Vis region, highlighting subtle spectral variations. Absorption peaks around the 600 nm wavelength appear to be associated with SOC content [2]. The absorption peak at 1100 nm is caused by the step in absorbance values at the junction of two spectrometer sensors [18]. At the 2050 nm wavelength, the depth of

Table 2

Model performance of GPR Based on different kernel functions for estimating SOC.

Training Set	R^2	RMSE [g / kg]	RPD	RPIQ
SE Kernel	0.8814	11.2773	2.9039	3.3465
RQ Kernel	0.9910	3.2739	10.5292	12.1339
Exponential Kernel	1.0000	0.0180	3850.7	4437.6
Matern Kernel	0.9833	4.0621	7.7429	8.9231
Hybrid Kernel	0.9719	5.2273	5.9700	6.8799
Test Set	R^2	RMSE [g / kg]	RPD	RPIQ
SE Kernel	0.7555	5.3324	2.0222	2.5169
RQ Kernel	0.7308	5.7818	1.9274	2.3989
Exponential Kernel	0.7359	5.7061	1.9460	2.4221
Matern Kernel	0.7584	5.3929	2.0344	2.5320
Hybrid Kernel	0.7671	5.2934	2.0721	2.5789

absorption increases with higher SOC content due to the relationship between nitrogen and organic matter Post and Noble [20]. The absorption feature at 2204 nm is typically caused by a combination of Al-OH bending and O-H stretching vibrations. This absorption characteristic is crucial for soils with low SOC content and is related to clay mineralogy.

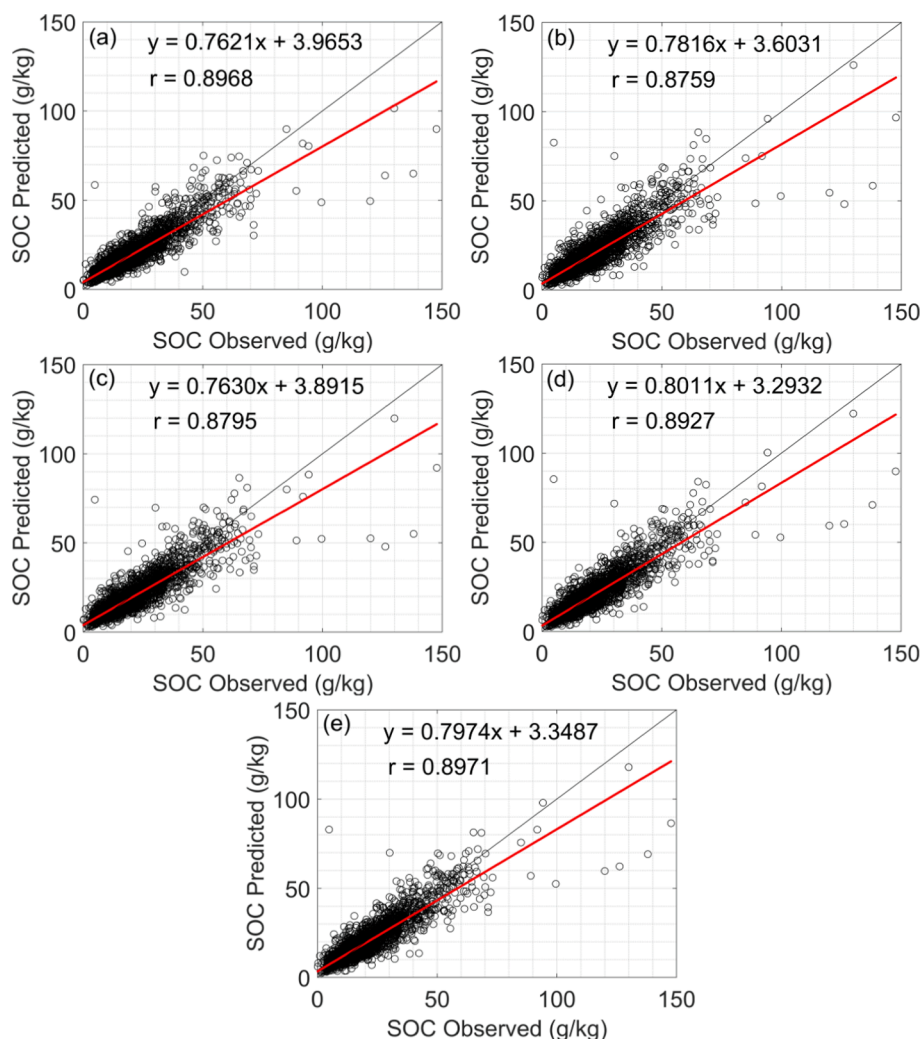


Fig. 3. Predicted versus observed values using GPR with different kernel functions: (a) SE kernel; (b) RQ kernel; (c) Exponential kernel; (d) Matern kernel; (e) Hybrid kernel. The letter 'r' represents the correlation coefficient between predicted and observed values. The red line represents the fitted line for the predicted values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Comparison of SOC estimation using different kernel functions

We estimated SOC using GPR models based on different kernel functions, with the model performance presented in Table 2. In the training set, a ten-fold cross-validation approach was used, and Bayesian optimization algorithms were applied to tune the hyperparameters of GPR models based on various kernel functions. After 30 iterations, the hyperparameters for the HKF-GPR model were determined to be: KernelParameters = $[-3.8388, 1.3544]$, Sigma = 0.1676. The Exponential kernel function achieved perfect model performance on the training set. However, its predictive performance on the test set was only marginally better than that of the RQ kernel, indicating that the Exponential kernel function exhibited the highest degree of overfitting and was unable to generalize effectively to the data in the validation set.

In the test set, the performance of all kernel functions naturally declined. The Hybrid kernel achieved the best predictive outcomes, with an R^2 of 0.7671, RMSE of 5.2934 g/kg, RPD of 2.0721, and RPIQ of 2.5789. Compared to GPR models based on single kernel functions, the HKF-GPR model exhibited superior SOC prediction performance, slightly improving all performance metrics. Due to limitations in computer hardware, the training time for a single model is excessively long, approximately seven days. Consequently, we did not test different weights, a and b , within the hybrid kernel function. Corresponding to the results in Table 2, the HKF-GPR model achieved the optimal fit, with

the highest correlation coefficient of 0.8972 between observed and predicted SOC values, as illustrated in Fig. 3.

Soil spectral data may contain a variety of complex signals and noise. Due to the characteristics of its function, the Exponential kernel might be overly sensitive to short-distance changes, capturing excessive noise. While this may appear beneficial in training data (achieving perfect fit), it actually reduces the model's adaptability to new data. In contrast, the SE kernel provides a smoother model that helps mitigate noise and outliers in the training data, thereby enhancing generalization capabilities. The Matern kernel produces functions that are rougher than those of the SE kernel and can capture more small-scale variations. Combining the SE and Matern kernels into a hybrid kernel allows for a more flexible representation of various features in the data, from smooth large-scale trends to rough local details. This combination offers a powerful approach to capturing the complex structures within soil hyperspectral data.

3.3. Comparison of SOC estimation using various regression models

In the training set, we utilized ten-fold cross-validation to determine the optimal number of latent variables for the PLSR model, setting it at 10. We also adjusted the hyperparameters for SVR and RF using ten-fold cross-validation combined with Bayesian optimization algorithms. After 30 iterations, the SVR hyperparameters were finalized as: Epsilon =

Table 3
Model performance of various regression methods for estimating SOC.

Training Set	R ²	RMSE [g / kg]	RPD	RPIQ
PLSR	0.6747	44.7991	1.7532	2.0204
SVR	0.7289	65.0828	1.9207	2.2135
RF	0.9471	9.9165	4.3476	5.0103
HKF-GPR	0.9719	5.2273	5.9700	6.8799
Test Set	R ²	RMSE [g / kg]	RPD	RPIQ
PLSR	0.4901	7.5959	1.4004	1.7430
SVR	0.5785	6.7727	1.5402	1.9170
RF	0.6257	6.7922	1.6346	2.0345
HKF-GPR	0.7671	5.2934	2.0721	2.5789

0.0007, KernelFunction = 'linear', KernelScale = 1, and Polynomial-Order = 0. For RF, the hyperparameters were set to: NumLearning-Cycles = 493, MinLeafSize = 2, MaxNumSplits = 4310, and NumVariablesToSample = 580.

As shown in Table 3, the HKF-GPR model consistently achieved the best model performance in both the training and test sets. The ranking of model performance from highest to lowest is as follows: HKF-GPR > RF > SVR > PLSR. In the test set, the RMSE of the RF model was 28.31 % higher, and its R² was 18.43 % lower than that of the HKF-GPR. Scatter plots of predicted versus observed SOC values for different regression models are depicted in Fig. 4. The scatter plot for the HKF-GPR model shows points closer to the 1:1 line, indicating higher accuracy. All models tended to underestimate samples with high SOC content.

PLSR may not effectively capture the complex nonlinear relationships present in large soil databases, which could explain its lower performance in both training and test sets. Both SVR and GPR are capable of handling nonlinear relationships effectively through the use of kernel functions. However, the performance of SVR is highly dependent on the selection of hyperparameters, such as the regularization coefficient, whereas GPR is a non-parametric model that does not require prior assumptions about the distribution of data. The RF model is

more sensitive to outliers and noise. Additionally, RF relies on simpler decision rules (tree splits), which may limit its performance when faced with the complex data structures present in large soil databases. Due to its probabilistic nature, GPR can flexibly adapt to noise and outliers in the data by adjusting the kernel function and noise parameters. GPR predictions also include confidence intervals, providing quantification of the model's uncertainty. Therefore, this may be the primary reason why the GPR model outperforms other regression models in estimating SOC within large soil databases.

Comparing the findings of this study with those of Ward et al. [28] and Liu et al. [12], Liu et al. [13], our results are superior to the earlier studies. Ward et al. [28] employed local PLSR to predict SOC in the LUCAS 2012 agricultural subset, achieving an R² of 0.67, RMSE of 5.16 g/kg, and RPD of 1.74. Our study's RMSE is slightly higher than that of Ward et al. [28], which can be attributed to the increase in prediction error of the spectral model as the standard deviation of the predicted soil properties increases [22]. Liu et al. [12], Liu et al. [13] combined feature selection algorithms and local PLSR to predict SOC content in the LUCAS 2015 agricultural subset, achieving an R² of 0.71, RMSE of 5.7 g/kg, and RPD of 1.87.

4. Conclusion

In this study, we developed various SOC estimation models using the Vis-NIR absorption spectra within the agricultural subset of the LUCAS 2015 soil database. We conducted a comparative analysis of the performance of different regression models and introduced a GPR model that utilizes a hybrid kernel function designed to adapt to the complex relationships present in large soil spectral databases. The results indicate:

The performance ranking of the regression models from highest to lowest is as follows: GPR > RF > SVR > PLSR.

Within the Gaussian regression process, the hybrid kernel function proposed in this paper demonstrates superior estimation accuracy

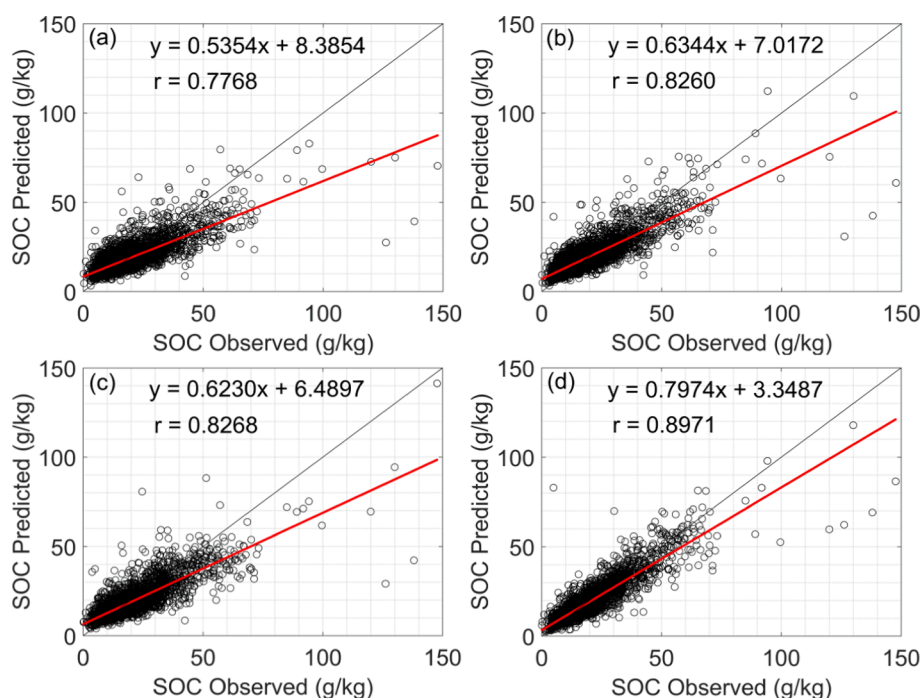


Fig. 4. Predicted versus observed values based on different regression models: (a) PLSR; (b) SVR; (c) RF; (d) HKF-GPR. The letter 'r' represents the correlation coefficient between predicted and observed values. The red line indicates the fitted line for predictions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

over single kernel functions, making the HKF-GPR model the most precise for SOC estimation.

However, the design of the hybrid kernel function in this study only considered several common kernel functions and did not account for the impact of hyperparameter prior distributions on the hybrid kernel. In the future, the design of the hybrid kernel will include other kernel functions, such as periodic and polynomial kernels, and incorporate hyperparameter prior distributions. In summary, GPR provides a more robust framework for handling the complexity of soil spectral data and SOC prediction in large databases, making it a promising option for such tasks.

CRediT authorship contribution statement

Baoyang Liu: Writing – original draft, Methodology, Formal analysis, Data curation. **Baofeng Guo:** Writing – review & editing, Project administration, Investigation, Funding acquisition. **Renxiong Zhuo:** Visualization, Conceptualization. **Fan Dai:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All the code, training set, test set, and validation set in this paper were publicly available on Github at https://github.com/Lbyyyyyyy/Hybrid-kernel-Gaussian-process-regression_Matlab.

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